🚀R&D DOCUMENT ON POINT-TO-SITE SETUP ON AZURE

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PURPOSE :
To research and demonstrate the step-by-step setup process of a Point-to-Site (P2S) VPN connection in Azure using certificate authentication.

🌐INTRODUCTION: POINT-TO-SITE VPN

- A point-to-site VPN is a connection between a single client machine and an Azure Virtual Network (VNet).
- It is ideal for remote users, developers, or testers.
- Requires no physical VPN device.
- Secure communication happens via certificates or Azure AD authentication.

🌐CERTIFICATION SETUP FOR AUTHENTICATION

Point-to-Site VPN in Azure supports certificate-based authentication, which uses a trusted root certificate and client certificates generated from it. This ensures only authorized users can establish VPN connections.

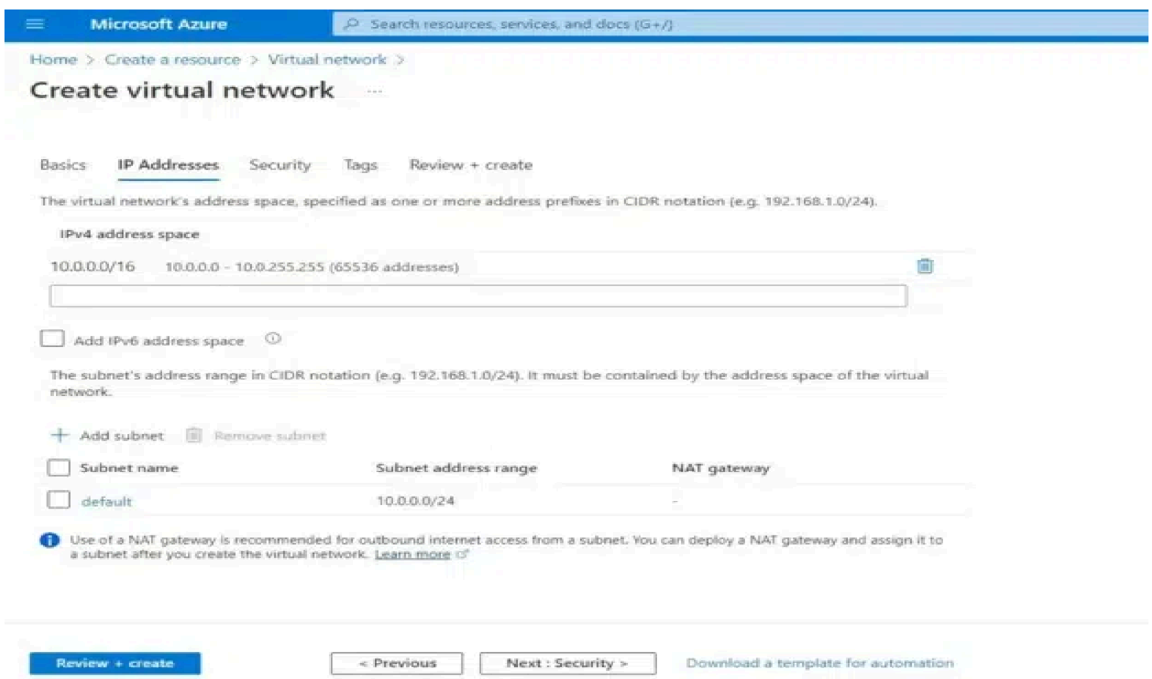
Required Certificates:

- Root Certificate: The trusted certificate that is uploaded to Azure VPN Gateway and used to validate client certs.
- Client Certificate: Installed on each client machine. Proves identity to Azure VPN Gateway.

🌐STEP-BY-STEP SETUP

1️⃣Create a Virtual Network

- Go to Azure Portal → Search “Virtual networks”
- Click Create
- Provide:
  - Name, Region
  - Address Space
  - Subnet
- Click Review + Create

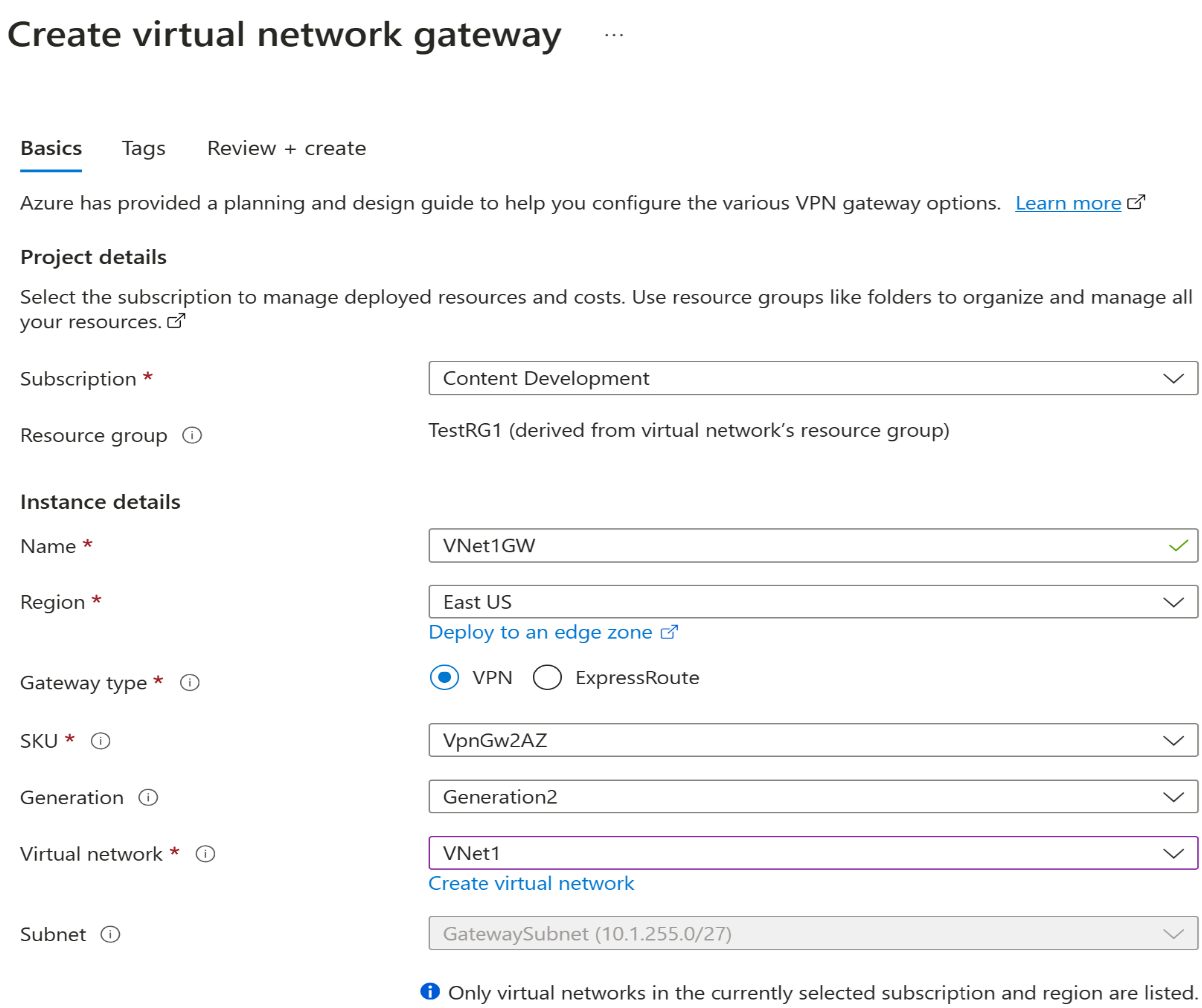


2️⃣ Create a Virtual Network Gateway

- Go to Azure Portal → Search "Virtual Network Gateways"
- Click Create
- Fill details:

- Name
- Region
- Gateway Type
- VPN Type
- SKU
- Virtual Network
- Public IP address

- Click Review + Create

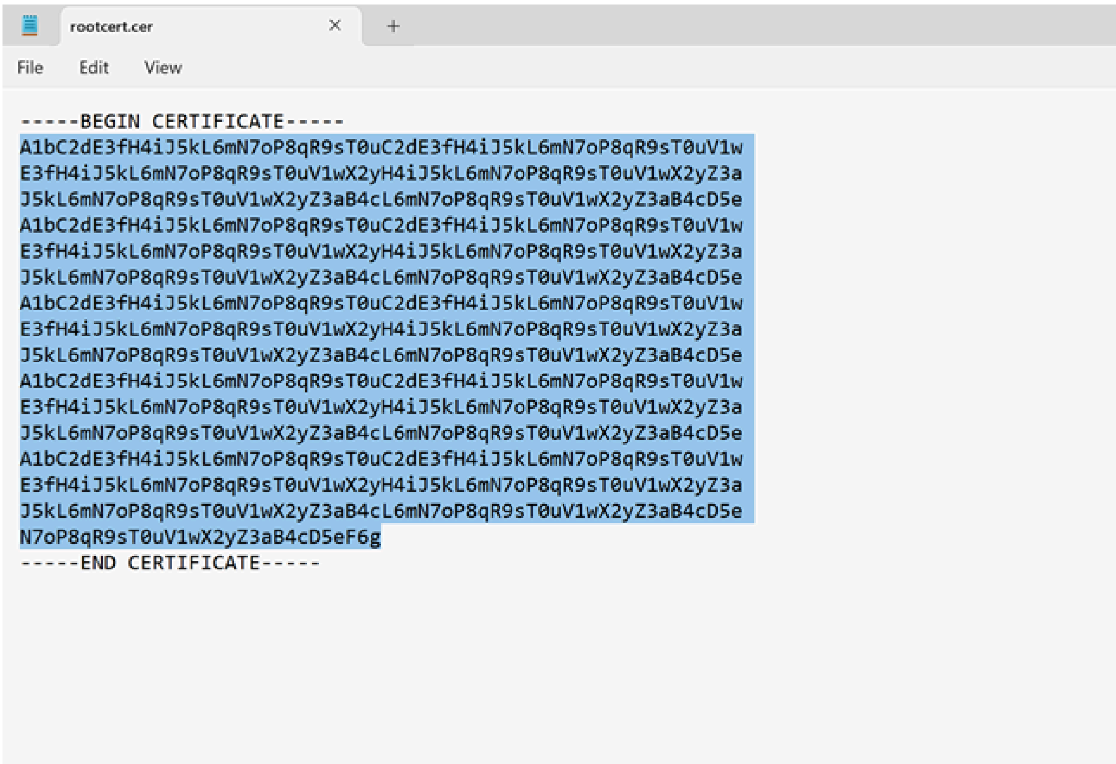


3️⃣ Generate Certificates for Authentication

a. Create a Root Certificate

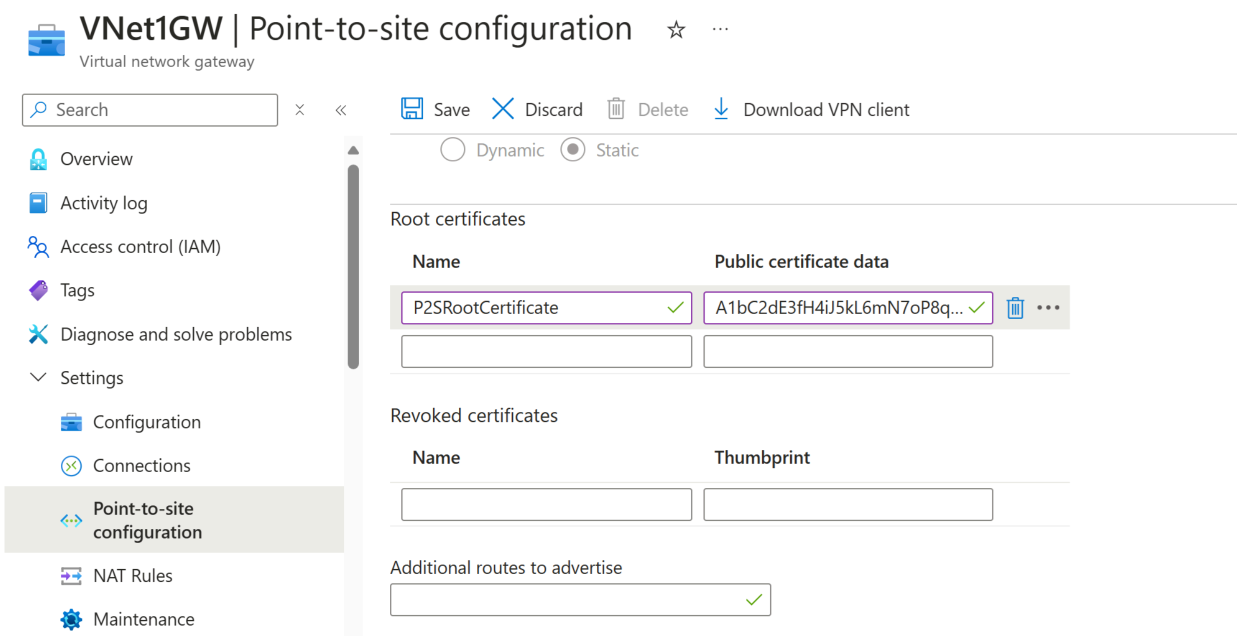
b. Export Public Key

c. Create a Client Certificate



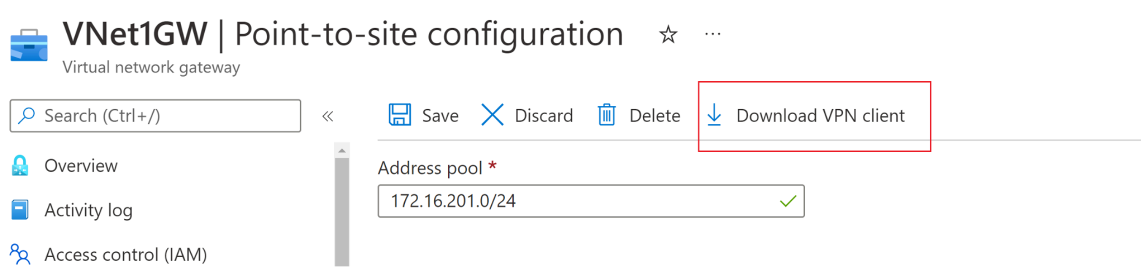
4️⃣ Configure Point-to-Site on the VPN Gateway

- Go to your VPN Gateway → Click Point-to-site configuration
- Click Configure now
- Fill:
    - Address Pool
    - Tunnel Type
    - Authentication Type
    - Root Certificate
- Click Save



5 Download VPN Client

- Go to the Point-to-Site configuration tab
- Click Download VPN client
- Choose the correct format based on your client OS (Windows, macOS, Linux)
- Extract the package and run the installer on the client machine



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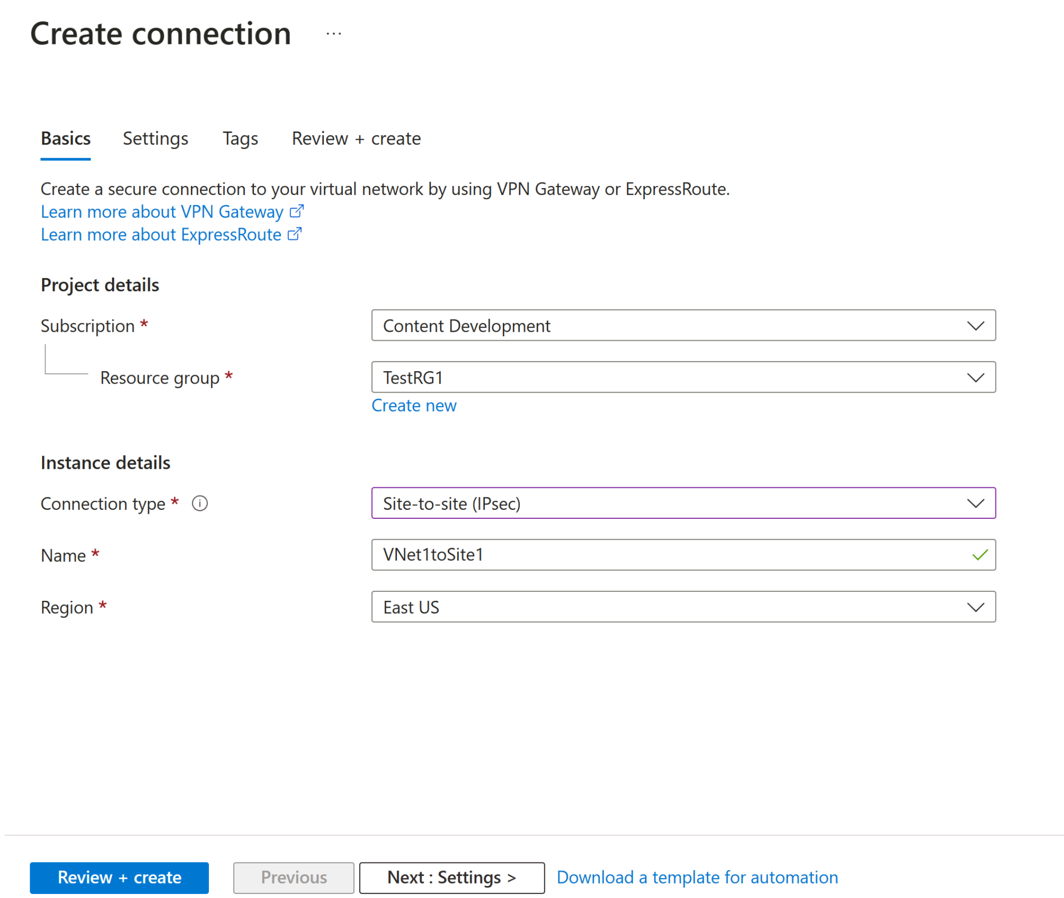
6 Connect to Azure via VPN

- On the client machine:
    - Go to Settings → Network & Internet → VPN
    - Find the installed VPN profile
    - Click Connect
    - You should now be securely connected to your Azure VNet



Verifying Connection

- Use ipconfig to verify the IP address from the VPN address pool.
- Use ping <VM-IP> or Remote Desktop to access resources in the VNet.
- Use Network Watcher or Connection Troubleshoot to debug issues.

⚙️ Tools Used

Tool	Purpose
Azure Portal	GUI to manage resources.

PowerShell	To generate and manage certificates
VPN Client	To connect from client machine
Network Watcher	To monitor and troubleshoot VPN

✅ Conclusion

Setting up a Point-to-Site VPN using certificate authentication is a secure and scalable solution for enabling remote users to access Azure resources without deploying dedicated VPN hardware.
